Innovation Network: Evidence from China listed Firms

Jingbo Cui[1] Te Lai[2]

[1]Duke Kunshan University; [2]Duke Kunshan University

August 21, 2021
AAERE

Motivation

- In business, win-win cooperation has become the most basic common sense. Both companies and individuals can increase their utility in trading and bring everyone into the busniess empire.
- Innovation is the foundation of a company's longevity. If a company wants to develop, it needs to better meet people's needs
- Although many papers have explored the impact of collaboration and innovation on firms, few studies have told us whether collaborative innovation brings greater benefits to firms
- In additon, We want to know if this network exists in China , and how has this network evolved.

Related Literature

- The burgeoning empirical literature in support for institutional investors' attention on
  - information disclosure of corporate environmental management and CSR
    - Cotter and Najah (2012), Dyck et al. (2019), Chen et al. (2020)
  - patent innovation
    - Aghion et al. (2013), Luong et al. (2017), Rong et al. (2017)
- The relevant literature has quantified the economic returns and a reduction in the risk of stock price collapse associated with CSR
  - Dyck et al. (2019), Wu and Hu (2019)

This Paper

- This paper studies the evolution of the patent networks and the the impact of the networks on the performance of the firms
- We visualize the network from 1996 to 2015 and the networks of some important industries
- Lastly,We not only look at the importance of these companies in pictures, but also quantify how important they are in the network

Data

- Chinese publicly-listed firms in stock markets of Shanghai and Shenzhen during the 1996-2015 period
  - Business data, like ROA, ROE, net asset, firm size, revenue and so on
- Patent applications are retrieved from Google Patent database
- listed firm network indicators
  - degree centrality, betweenness centrality, closeness centrality

Variable Construction

- Dependent variables are return of asset and return of equity
  - average return of asset (ROAA), average return of equity (ROEA), weighted return of asset (ROAE), weighted return of equity (ROEE)
- Independent variables are degree centrality, betweenness centrality, closeness centrality
- Control variables: year, industry and firm indicators

Empirical Model

- We propose a model to account for the role of industry pollution intensity while estimating the impacts of institutional investors on firm-level innovation direction

$$\begin{aligned} EnvInnovRatio_{it} =& \alpha_0 + \alpha_1 InstInvest_{it-1} \times Dirty_j + \alpha_2 InstInvest_{it-1} \\ &+ \beta X_{it-1} + \lambda_i + \delta_{rt} + \mu_{jt} + \epsilon_{ijrt}, \end{aligned} \tag{1}$$

where

  - $EnvInnovRatio_{it}$: innovation direction
  - $InstInvest_{it-1}$: number of shareholdings owned by institutional investors
  - $Dirty_j$: time-invariant industry pollution intensity
  - X_{it-1}: a set of firm-level attributes
  - λ_i: firm-level fixed effect
  - δ_{rt} and μ_{jt}: regional-year and industry-year fixed effects or linear trend
  - ϵ_{ijrt}: disturbance term

Baseline Results

VARIABLES	Baseline OLS Method				IV Method	
					1st stage	2nd stage
	(1)	(2)	(3)	(4)	(5)	(6)
HS300					0.044***	
					(0.005)	
InstInvest	-0.038*	-0.037*	-0.032	-0.032		0.015
	(0.020)	(0.020)	(0.020)	(0.020)		(0.080)
InstInvest × SO2	0.717***	0.664***	0.690**	0.631***		2.137**
	(0.249)	(0.214)	(0.259)	(0.222)		(0.897)
Observations	6,179	6,179	6,179	6,179	6,179	6,179
R-squared	0.005	0.011	0.013	0.020	0.050	0.039
Firm Attributes	Y	Y	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
Province linear trend		Y		Y	Y	Y
Industry linear trend			Y	Y	Y	Y

Robustness Checks

VARIABLES	City Confounders (1)	One-year Forward Patent (2)	Alternative Dirty Industry (3)	EnvrPatent (4)
InstInvest	-0.042**	-0.034	-0.02	-0.083
	(0.020)	(0.031)	(0.015)	(0.095)
InstInvest × SO2	0.861***	0.458*		3.161**
	(0.215)	(0.238)		(1.160)
InstInvest × Coal			0.518***	
			(0.101)	
Observations	6,179	5,392	6,076	6,179
R-squared	0.06	0.032	0.019	0.105
Firm Attributes	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y
Province linear trend		Y	Y	Y
Industry linear trend	Y	Y	Y	Y
City linear trend	Y			

Heterogeneity by Patent Type and Inst Investor Type

VARIABLES	EnvrPatent Ratio				
	Invention	Utility	All	Invention	Utility
	(1)	(2)	(3)	(4)	(5)
InstInvest	-0.026*	-0.013			
	(0.015)	(0.018)			
InstInvest × SO2	0.519***	0.367**			
	(0.178)	(0.165)			
DomInstInvest			-0.031	-0.028*	-0.010
			(0.020)	(0.015)	(0.019)
DomInstInvest × SO2			0.624***	0.536***	0.340**
			(0.220)	(0.169)	(0.167)
ForInstInvest			-0.110	0.120	-0.216*
			(0.098)	(0.163)	(0.116)
ForInstInvest × SO2			1.002	-0.563	2.133
			(1.221)	(1.594)	(1.660)
Observations	6,179	6,179	6,179	6,179	6,179
R-squared	0.016	0.022	0.02	0.016	0.022
Firm Attributes	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y	Y
Province linear trend	Y	Y	Y	Y	Y

Financial Constraints Mechanism

VARIABLES	Tight Financial Constraints $KZ = 1$		Soft Financial Constraints $KZ = 0$	
	(1)	(2)	(3)	(4)
InstInvest	-0.043		-0.008	
	(0.033)		(0.016)	
InstInvest × SO2	0.992***		-0.358	
	(0.265)		(0.271)	
DomInstInvest		-0.045		-0.004
		(0.033)		(0.016)
DomInstInvest × SO2		1.007***		-0.449
		(0.270)		(0.333)
ForInstInvest		0.066		-0.317***
		(0.144)		(0.047)
ForInstInvest × SO2		0.214		5.912*
		(1.482)		(3.320)
Observations	3,595	3,595	2,584	2,584
R-squared	0.036	0.036	0.037	0.038
Firm Attributes	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y
Province linear trend	Y	Y	Y	Y

Corporate Governance Mechanism

VARIABLES	Strong Corporate Governance $ZL = 1$		Weak Corporate Governance $ZL = 0$	
	(1)	(2)	(3)	(4)
InstInvest	0.058		-0.040	
	(0.044)		(0.024)	
InstInvest × SO2	0.178		0.566**	
	(0.683)		(0.251)	
DomInstInvest		0.060		-0.038
		(0.046)		(0.024)
DomInstInvest × SO2		0.049		0.572**
		(0.672)		(0.255)
ForInstInvest		-0.064		-0.133
		(0.447)		(0.124)
ForInstInvest × SO2		11.654***		-0.065
		(1.805)		(1.059)
Observations	966	966	5,213	5,213
R-squared	0.105	0.108	0.021	0.022
Firm Attributes	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y
Province linear trend	Y	Y	Y	Y

Market Value Mechanism

VARIABLES	Revenue		TobinQ		ROA	
	(1)	(2)	(3)	(4)	(5)	(6)
InstInvest×SO2	-0.004		0.077		-0.004	
	(0.057)		(0.049)		(0.057)	
InstInvest×SO2×EnvrPatent	0.011**		0.007***		0.011**	
	(0.005)		(0.002)		(0.005)	
DomInstInvest×SO2		0.002		0.081		0.002
		(0.057)		(0.049)		(0.057)
DomInstInvest×SO2×EnvrPatent		0.011**		0.006***		0.011**
		(0.005)		(0.002)		(0.005)
ForInstInvest×SO2		-0.420		-0.165*		-0.420
		(0.306)		(0.094)		(0.306)
ForInstInvest×SO2×EnvrPatent		0.172**		0.127		0.172**
		(0.080)		(0.076)		(0.080)
Observations	5,160	5,160	5,452	5,452	5,160	5,160
R-squared	0.234	0.234	0.329	0.33	0.135	0.135
Firm Attributes	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y	Y	Y
Province linear trend	Y	Y	Y	Y	Y	Y
Industry linear trend	Y	Y	Y	Y	Y	Y

Takeaway Points

- Institutional investors' shareholding ratios are positively correlated with the environmental innovation direction of firms
  - pollution-intensive sectors vs. non-pollution-intensive sectors
  - robust against a series of alternative checks.
- Some heterogeneity results arise
  - domestic vs. foreign institutional investors
- We reveal the roles of institutional investors in corporate innovation direction
  - the financial easing role
  - the role of corporate governance
  - institutional investors' incentives are aligned with corporates

Thank You

- Cui, Jingbo, Shaozhou Qi, Jia Xu, and Shu Zeng, 2021, "Institutional Investors and Corporate Environmental Innovation," SSRN Working Paper
- For further comments, correspondence may be sent to
  - Jingbo Cui, Duke Kunshan University
    - email: jingbo.cui@duke.edu
    - personal website: http://sites.google.com/site/jbcui2013